

Final Year Project

PROPOSAL



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Abstract

Flight delays remain a persistent challenge in the aviation industry, causing financial losses for airlines, disrupting flight operations, and frustrating passengers. Existing delay management approaches depend largely on fixed schedules or manual decision-making, which are insufficient for handling the complexity of modern air traffic. Variables such as adverse weather, airport congestion, aircraft rotation, and crew availability create dynamic conditions that make delay forecasting difficult. This lack of accurate predictive systems leads to inefficiencies, reduced passenger trust, and operational strain on flight managers. Addressing this challenge is vital to improve airline efficiency, reduce costs, and enhance the overall passenger experience.

This project, Skylytics – An AI Powered Flight Delay Prediction System, proposes an intelligent solution that integrates multiple data sources including historical flight records, real-time weather data, and airport information. Using a hybrid machine learning approach combining models such as XGBoost and LSTM/Transformer, the system will forecast both the likelihood and duration of delays. Explainable AI techniques (e.g., SHAP) will provide transparency by highlighting the key factors influencing each prediction. A web dashboard (Next.js, Tailwind) and an iOS mobile application (SwiftUI) will deliver user-friendly interfaces for both flight managers and passengers, offering features like what-if analysis, delay heatmaps, smart notifications, and an AI assistant. Cloud deployment with FastAPI and scalable databases will ensure fast, reliable, and professional performance, making Skylytics a practical and industry-relevant Final Year Project.

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1. Introduction

Air travel plays a vital role in modern global connectivity, supporting the movement of people, goods, and services while driving economic growth and cultural exchange. With millions of passengers depending on airlines daily, reliability and punctuality are critical. However, flight delays remain one of the most persistent challenges in aviation, creating cascading disruptions, financial losses, and passenger dissatisfaction. Airlines operate on tight schedules where a minor delay can ripple across multiple flights, while passengers often face missed connections, business losses, or added expenses.

Globally, billions of dollars are lost annually due to delays. In the United States alone, thousands of flights are delayed each year by more than 15 minutes, as reported by the Bureau of Transportation Statistics (BTS). Causes range from adverse weather and air traffic congestion to crew scheduling and technical issues. These complexities make delay prediction a difficult but crucial task for improving aviation reliability. Traditional systems relying on manual assessments or static schedules are inadequate, as they fail to process the vast, real-time data required to capture non-linear and interdependent delay factors.

Artificial Intelligence (AI) and Machine Learning (ML) provide a transformative solution. By leveraging massive datasets - including historical flight records, weather data, and airport congestion - AI models can detect hidden patterns and generate real-time predictions. This empowers airline operation managers to proactively address disruptions and enhances passengers' trust in airline services.

Skylytics is proposed as an AI-powered flight delay prediction system designed to address these challenges. By integrating data from BTS, OpenFlights, and Meteostat, the system will employ hybrid ML models (XGBoost, LSTMs, and Transformers) to predict both delay likelihood and duration. Beyond prediction, Skylytics emphasizes explainability and usability: managers will gain insights into “why” a flight is delayed through SHAP values, test alternate scenarios with what-if simulations, and collaborate with teams using built-in communication tools. Passengers, in turn, will access real-time updates, delay notifications, and monitoring features.

This project represents more than delay forecasting - it envisions a holistic aviation management ecosystem. By fusing predictive analytics, explainable AI, and user-focused design, Skylytics aims to reduce uncertainty, optimize airline operations, and enhance passenger experience. For this Final Year Project (FYP), it offers both deep technical learning and real-world impact, setting a benchmark for applying AI to practical aviation challenges.

2. Problem Description

2. 1 Primary Scope

The aviation industry faces one of its most persistent challenges in the form of flight delays, which impact both airline operators and passengers. The problem is not merely about late arrivals; it extends to cascading operational inefficiencies, increased financial costs, and significant passenger dissatisfaction. Despite decades of technological advancements, most existing delay management systems rely on static schedules, manual monitoring, or limited statistical models that are insufficient for capturing the dynamic and multi-factor nature of delays.

Flight delays can be caused by diverse factors such as adverse weather, airport congestion, technical malfunctions, crew scheduling conflicts, and even cascading effects from earlier delayed flights. Each of these components interacts in non-linear ways, making manual prediction nearly impossible. Without reliable predictive tools, airline managers are left to reactively manage crises rather than proactively preventing them.

This project seeks to answer three fundamental questions:

- What is the problem?

Persistent and costly flight delays that reduce reliability in aviation.

- Why is it important?

Because delays harm airlines financially, frustrate passengers, and reduce overall trust in air travel systems.

- How can it be solved?

By leveraging AI-driven predictive analytics, historical and real-time datasets, and explainable machine learning models to forecast flight delays accurately and provide actionable insights.

By building an intelligent predictive system, this project addresses a pressing real-world challenge while contributing innovative solutions that align with global trends toward AI-enabled decision support systems in aviation.

2. 2 Final Deliverables and Beneficiaries

The product of this work will be a web-based platform and iOS application powered by an AI-driven backend capable of predicting flight delays and providing actionable insights.

- **Direct beneficiaries:** Airline operation managers, flight schedules, and ground staff who can make data-driven decisions to reduce operational inefficiencies.
- **Indirect beneficiaries:** Passengers, who will receive more reliable updates and improved travel experiences, and airlines, which will reduce costs and protect their reputation.

2.3 Optional Scope

If resources allow, additional premium integrations will be considered, such as real-time aviation data APIs (e. g., FlightAware, FlightRadar24) for higher prediction accuracy, and 3D flight visualization modules (via Three.js) to enhance user interactivity. While not mandatory, these features would significantly expand the usability and professional depth of the system.

2.4 Objectives

1. The primary objective of this project is to develop a reliable and industry-ready flight delay prediction system that integrates multiple data sources such as flight schedules, weather information, and airport operational records.
2. For the prototype phase, tree-based models such as XGBoost, Random Forest, and LightGBM will be used to establish the baseline system.
3. In the advanced phase, the system will move to hybrid ensemble models that combine XGBoost with LSTM and Transformer architectures to predict both the likelihood and the duration of delays.
4. Explainable AI techniques such as SHAP and LIME will be incorporated to ensure transparency and trust, allowing operational managers to clearly understand the factors influencing each prediction.
5. The solution will include a professional and user-friendly interface in the form of a web-based dashboard, featuring interactive capabilities such as delay heatmaps, what-if analysis, and smart notifications.
6. An iOS mobile application will also be developed to provide quick access to predictions and real-time monitoring for operational managers and staff.
7. Performance optimization will be achieved through caching, load balancing, and efficient cloud deployment strategies, with hosting support provided by platforms such as AWS, GCP, and Vercel, along with Apple Developer accounts for iOS distribution.
8. The system will be rigorously evaluated using established accuracy metrics such as ROC-AUC, RMSE, and F1-score, and complemented with user experience testing to validate its effectiveness in real-world operational scenarios.

2.5 Novelty

The novelty of this project lies in its hybrid modeling approach (XGBoost combined with LSTM/Transformer), explainable predictions through SHAP-based feature importance visualization, and multi-platform usability with both a professional web dashboard and an iOS app. It further stands out with advanced interactive features such as what-if analysis, delay heatmaps, and collaboration tools, alongside professional deployment on cloud platforms with SEO-friendly, responsive, and scalable design.

3. Literature Review

3.1 Flight Delay Prediction

Flight delay prediction remains one of the most persistent challenges in the aviation industry. According to the U. S. Bureau of Transportation Statistics (BTS), nearly 20% of all flights experience delays annually, creating cascading effects for airlines, airports, and passengers [1]. Traditional methods of prediction relied heavily on regression models or historical averages [2]. These statistical tools, though useful for broad patterns, failed to capture the dynamic, multi-factorial causes of delays such as weather, airport congestion, and aircraft rotations.

With the increasing availability of large aviation datasets, researchers have shifted towards machine learning approaches. Rebollo and Balakrishnan [3] highlighted the insufficiency of static tools in fast-changing traffic conditions and advocated for data-driven models. These approaches allow more accurate and flexible prediction by incorporating diverse data sources.

3.2 Machine Learning Models Used in Delay Prediction

Several ML models have been applied in flight delay prediction:

- **Decision Trees and Random Forest (RF):**
Random Forest has been widely adopted due to its interpretability and robustness. Xu et al. [4] demonstrated its strength in managing heterogeneous data types and nonlinear interactions in flight operations.
- **Gradient Boosting Models (XGBoost, LightGBM):**
Chen and Guestrin [5] introduced XGBoost, which has since become a standard in aviation prediction tasks. It efficiently handles missing values and complex interactions, making it suitable for binary tasks like predicting whether a flight will be delayed.
- **Deep Learning (LSTM, Transformers):**
LSTM models are particularly strong at handling sequential data, making them effective in modeling delay propagation from inbound flights [6]. More recently, Transformer-based

approaches have been applied for aviation time-series forecasting, handling long dependencies with superior performance [7].

- **Hybrid Approaches:** Combining ensemble models (like XGBoost) with temporal models (LSTM/Transformer) improves prediction performance. Chen and Huang [8] showed that hybrid systems consistently outperform individual models, though they require more computational resources.

3. 3 Explainability in AI for Aviation

While machine learning models can achieve high accuracy, their adoption in aviation requires trust and transparency. Black-box systems are often resisted by airlines and regulators. Explainable AI (XAI) frameworks such as SHAP (SHapley Additive Explanations) and LIME provide insights into feature contributions [9].

In aviation delay prediction, this means managers can understand whether weather, traffic, or operational factors are most responsible for a predicted delay. While explainability has been widely explored in fields like finance and healthcare [10], its systematic use in aviation systems is still limited. Zhang et al. [12] demonstrated that providing feature attribution significantly improved operator confidence in delay prediction tools.

3. 4 Gaps in Existing Research

From current literature, the following gaps are clear:

1. **Lack of real-time data integration:** Many models rely solely on static datasets and ignore live updates [11].
2. **Low practical deployment:** Few studies develop production-ready systems accessible to airlines.
3. **Explainability gap:** Accuracy is prioritized over interpretability, limiting adoption [12].
4. **Passenger-facing systems are rare:** Most research is operations-focused.
5. **Scalability challenges:** Scaling deep models to global datasets remains a bottleneck [13].
6. **Simulation and what-if analysis:** Almost absent in the reviewed literature [14].

3. 5 Summary

The reviewed literature confirms the potential of machine learning in aviation but highlights critical shortcomings. *Skylytics* addresses these gaps by:

- Using a **hybrid ML model** (XGBoost + LSTM/Transformer).

- Integrating **real-time data streams** (weather + operations).
- Applying **XAI techniques** (SHAP, LIME) to ensure transparency.
- Building both **manager- and passenger-facing interfaces**.
- Providing **simulation, history tracking, and performance analytics**.

This dual focus on academic rigor and practical deployment ensures novelty and real-world impact.

3. 6 References

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4. Methodology

The methodology of this project is structured as an iterative, end-to-end workflow that begins with data collection and preparation, advances into model development and optimization, and concludes

with evaluation, deployment, and user-facing application design. To ensure technical soundness and practical usability, the process is divided into two phases: prototyping (establishing feasibility and baselines) and advanced development (building scalable, explainable, and production-ready models).

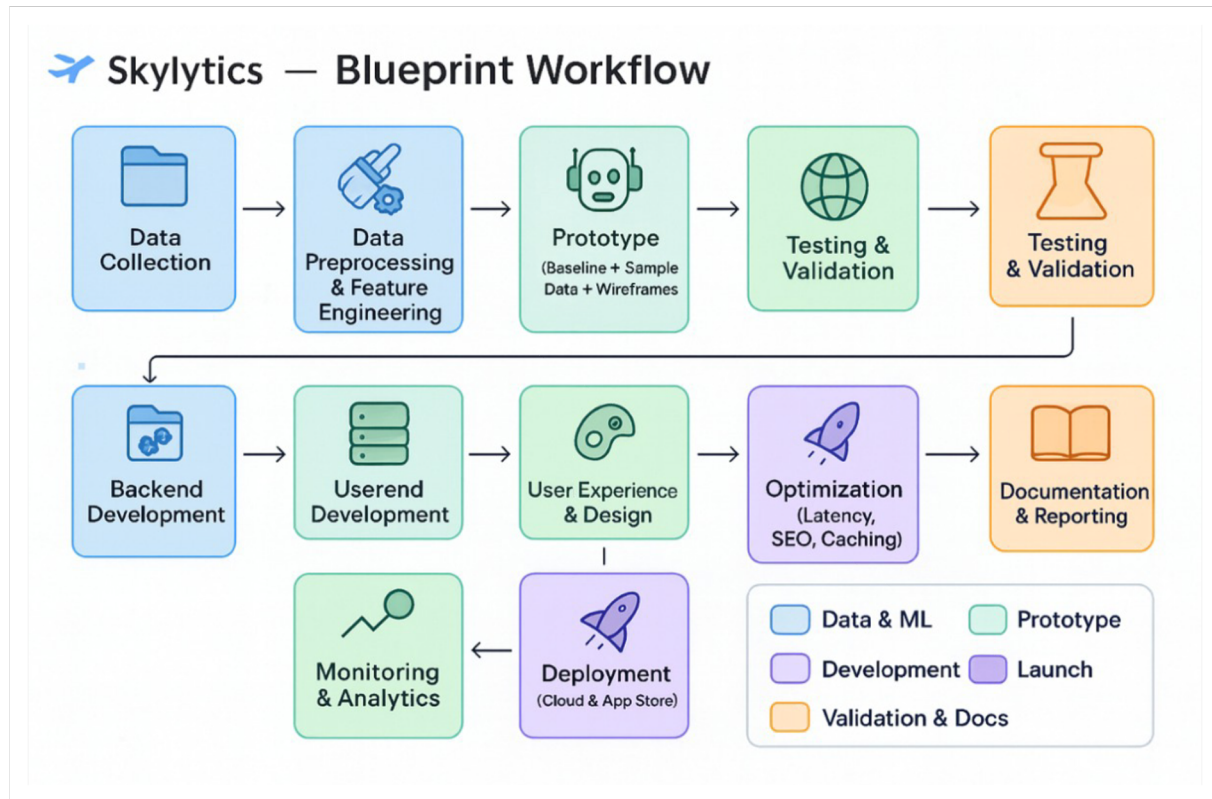


Figure 1

4. 1 Prototyping Phase

The prototyping phase establishes the foundation of the system through simplified datasets and baseline models. This stage validates feasibility, sets benchmark metrics, and highlights potential challenges before investing in complex architectures.

- **Dataset Selection and Preprocessing**

Historical flight delay records from Kaggle and BTS form the initial dataset. Raw data is cleaned to address missing values, standardize time formats, and ensure logical consistency between scheduled and actual flights. Outliers are retained where they represent realistic events, such as severe weather.

- **Baseline Modeling**

Simple models such as Logistic Regression and Decision Trees are implemented to measure

predictive potential. These benchmarks serve as a reference for evaluating improvements gained through more advanced models later.

- **Pipeline Validation**

The prototype includes exploratory data analysis (EDA), basic ETL scripts, and small-scale experiments in scikit-learn. This confirms that the system can process aviation datasets reliably and lays the groundwork for more complex workflows.

4.2 Advanced Model Development Phase

Building on the validated prototype, this phase introduces advanced models, feature engineering, and system integration for real-world application.

- **Data Handling and Feature Engineering**

Multiple data sources including Meteostat (weather) and OpenFlights (airport routes) are integrated. Feature engineering introduces derived attributes such as congestion indices, day-of-week encodings, and historical averages. Time-based features are encoded cyclically, categorical variables are machine-encoded, and scaling ensures consistency.

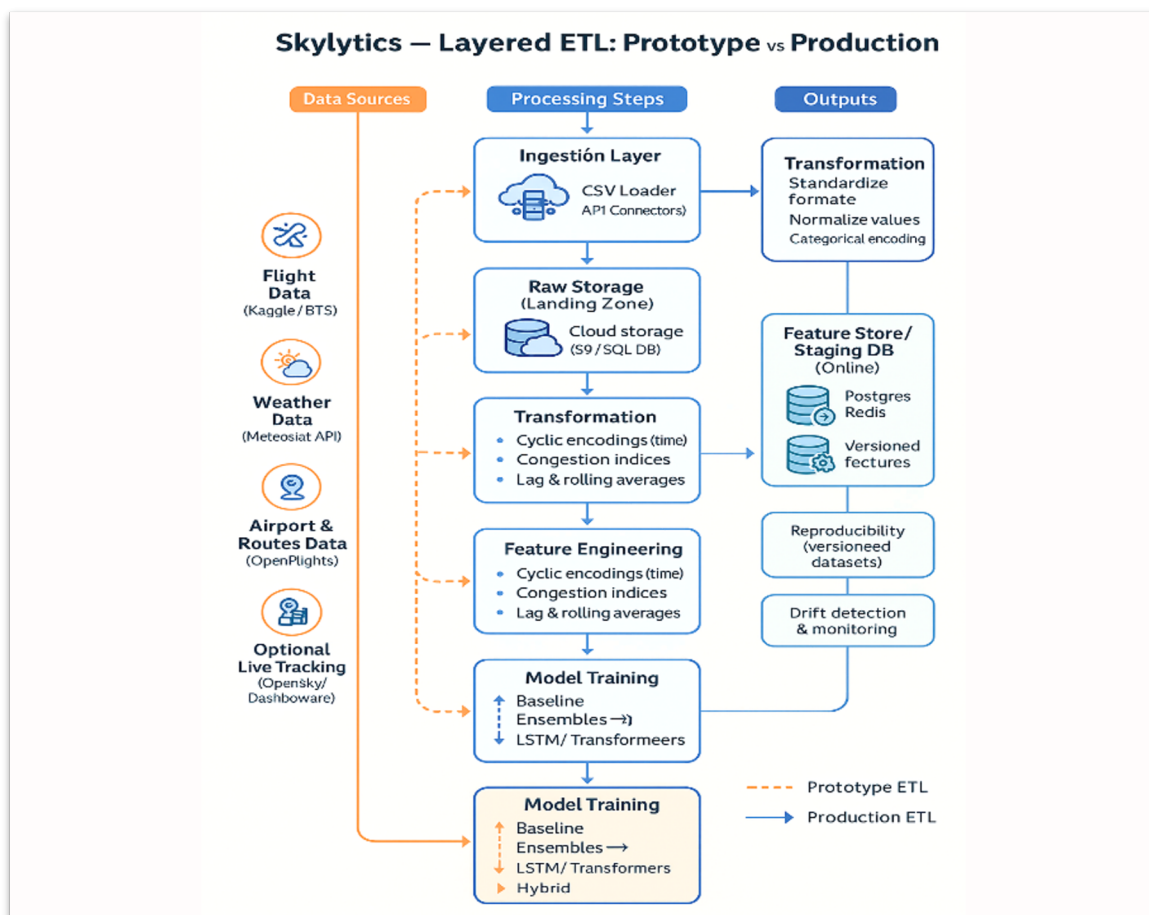


Figure 2

- **Ensemble and Sequential Models**

Gradient boosting models (XGBoost, LightGBM) capture nonlinear interactions. To address sequential dependencies, Long Short-Term Memory (LSTM) networks and Transformers are incorporated. Outputs from these models are fused into a hybrid ensemble, combining predictive accuracy with interpretability.

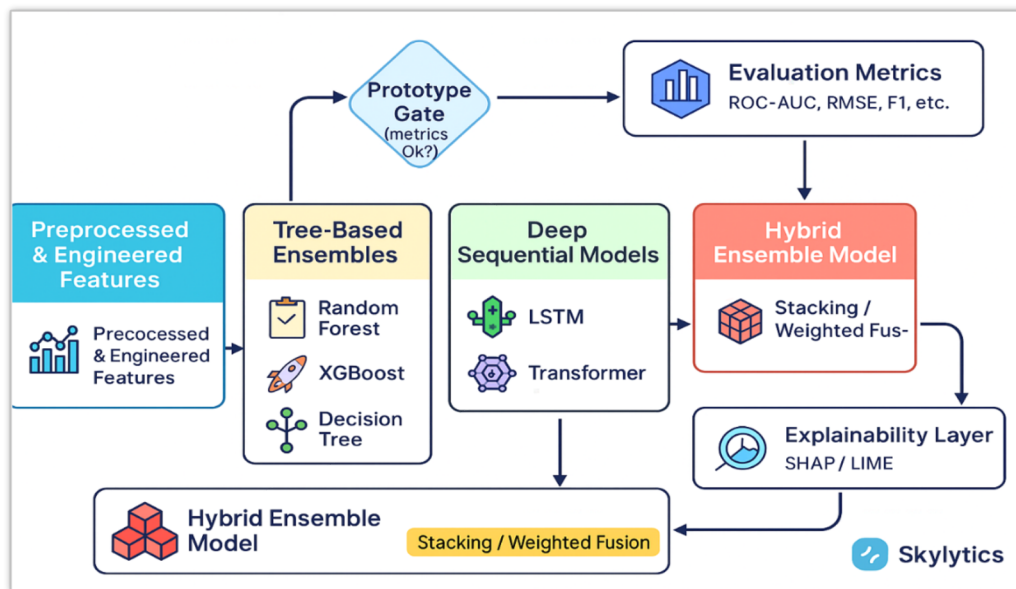


Figure 3

- **Optimization and Validation**

Hyperparameter tuning (Randomized Search, Optuna) ensures performance efficiency. Cross-validation mitigates overfitting, while imbalanced classes are addressed through SMOTE and class reweighting.

4.3 Evaluation, Explainability, and Deployment

Once trained, models are evaluated with multiple metrics, since accuracy alone is insufficient in imbalanced datasets. Precision, Recall, F1-Score, and ROC-AUC provide a multi-dimensional performance view. Misclassifications undergo error analysis to refine model design.

Explainability tools such as SHAP and LIME ensure transparency by attributing predictions to features like weather or congestion. This improves trust and enables operational managers to understand model behavior.

For deployment, FastAPI serves predictions through real-time APIs, with Docker containers ensuring portability. Redis provides caching for fast queries, and optional Kubernetes orchestration supports scalability. Monitoring pipelines detect data drift to maintain reliability.

4.4 User Interfaces and Applications

The final stage emphasizes usability.

- **Web Dashboard (Next.js + Tailwind)** for airline managers, featuring delay predictions, SHAP-based explanations, delay heatmaps, and what-if analysis tools. Collaboration features allow teams to share insights and coordinate responses.
- **iOS Mobile App (SwiftUI)** for passengers, focusing on real-time predictions, monitoring, and push notifications. Both interfaces connect securely to the backend via APIs, ensuring consistency in results.

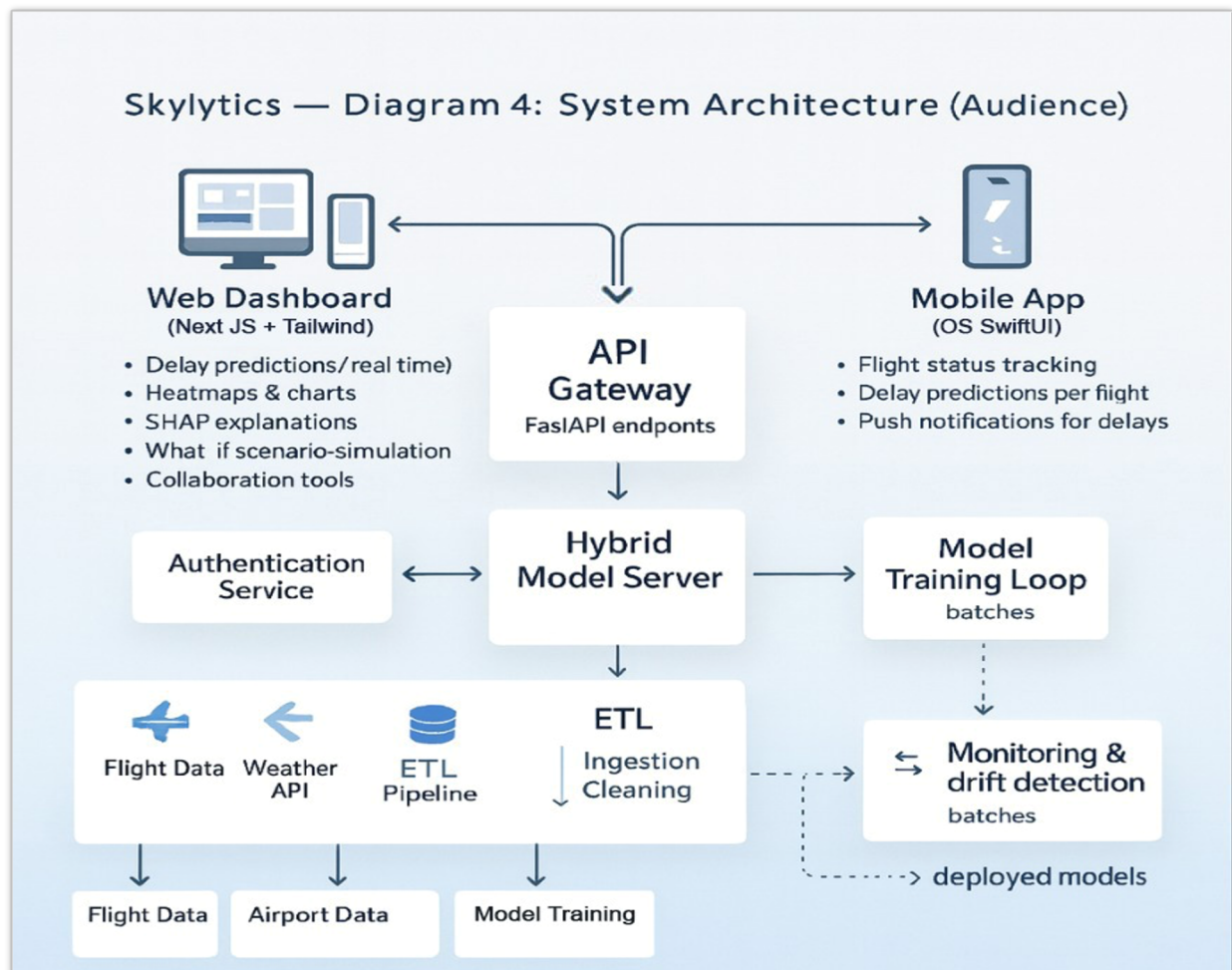


Figure 4

4.5 Workflow Summary

The methodology follows a two-stage evolution:

1. **Prototyping** with historical datasets and baseline models to validate feasibility and set benchmarks.
2. **Advanced Development** with integrated datasets, hybrid models, explainability, and production deployment for end-user applications.

This staged approach ensures that the system matures from a feasible prototype into a robust, industry-ready solution that balances technical rigor, scalability, and usability.

5. Feasibility Plan

The feasibility of Skylytics is justified through available resources, the expertise of the team, the choice of tools and technologies, and a realistic budget. The project is designed to remain achievable within the given timeframe while offering technical depth and practical value.

5.1 Resource Requirement

The project requires computing power for data processing, model training, and system deployment. Standard laptops with cloud-based support (Google Colab/AWS credits) will be sufficient. Additional requirements include real-time APIs for weather and flight data, database hosting, and deployment platforms. No specialized hardware is required.

5.1.1 Expertise of the Team

The team collectively has the right foundation and interest for successful completion:

- **Abdul Mutlib Ali:** Data Science/ML - Python, Kaggle, and scikit-learn for baseline-to-improved models.

- **Abuzar Rashid:** Web & Integration - Node/Express, REST APIs, and relational databases.

- **Kumail Ahmed:** iOS & Frontend - SwiftUI app with API integration; basic React/Next.js. All members have studied Databases, Machine Learning, Networking, and Software Engineering, ensuring preparation. The project aligns with our interests and offers a focused, balanced learning curve.

5. 1. 2 Tools / Technology

The tools and technologies required are open-source or accessible through educational resources:

Backend: FastAPI (lightweight, asynchronous, Python-based), Docker for deployment.

- **Frontend:** Next.js (SEO-friendly React framework), Tailwind CSS, Recharts/Plotly for visualizations.
- **Mobile App:** SwiftUI for native iOS development.
- **Database:** PostgreSQL (relational storage) and Redis (caching).
- **Machine Learning:** Python (Pandas, Scikit-learn, XGBoost, PyTorch).
- **Data Sources/APIs:**
 1. Kaggle BTS Flight Delay Dataset (2015) – <https://www.kaggle.com/datasets/usdot/flight-delays?resource=download>
 2. Meteostat API for weather – <https://meteostat.net>
 3. OpenFlights (airports, routes) – <https://openflights.org/data.html>
 4. Optional: OpenSky Network for live traffic – <https://opensky-network.org>
- **Hosting/Deployment:** AWS EC2 (backend), AWS RDS (database), Vercel (frontend), Apple Developer account (iOS).

5. 1. 3 Budget

The project has minimal financial requirements. Most tools and datasets are free. Costs include:

- **Apple Developer Account:** \$99/year.
- **Professional Hosting (AWS EC2 + RDS):** \$10–15/month.

Total estimated cost: **under PKR 90,000**, which is manageable.

5. 2 Risks Involved

The proposed project involves some potential risks that need to be considered:

1. **Data Availability:** Real-time APIs (e.g., Meteostat, OpenSky) may have usage limits or downtime.

Mitigation: Use cached data and maintain fallback Kaggle datasets for continuity.

2. **Model Accuracy:** Flight delay prediction is inherently uncertain due to unpredictable events (e.g., emergencies).

Mitigation: Use hybrid models and explainable AI to improve trust and reliability.

3. **Compute Limitations:** Training advanced models may require high processing power.

Mitigation: Use cloud services (Google Colab Pro, AWS) for scalability.

4. **Integration Complexity:** Multi-platform deployment (web + iOS) may create integration challenges.

Mitigation: Follow modular development with API-first design.

Overall, these risks are manageable and do not hinder the successful completion of the project.

6. Key Milestones and Schedule

6.1 Key Milestones

Following are the major milestones of the project and their related information.

Table 0-1: Breakdown of work in form of milestones

S. No.	S. No. of Predecessor	Milestone Key Milestone Name / Description	Duration (person-hours)
1	-	Prototyping Data Setup (Kaggle, Meteostat, OpenFlights)	81
2	1	Baseline Prototype Model (XGBoost classifier & regressor)	81
3	2	Prototype Frontend (Basic Web UI for predictions)	54
4	-	Production Data Collection & API Integration (BTS, Meteostat live)	81
5	4	Advanced Model – Phase 1 (LSTM/Transformer & Graph POC)	54
6	-	Contingency / Buffer (EDA, fixes, reviews)	81
7	5	Advanced Model – Phase 2 (Hybrid integration & tuning)	162
8	7	Backend API Development (FastAPI, Redis caching, PostgreSQL)	72
9	7, 8	Full Web Frontend (Next.js + Tailwind + Explainable AI UI)	72
10	8	iOS App Development (SwiftUI + notifications, saved flights)	36
11	7-10	System Integration & Internal Testing	30
12	11	Hosting & Deployment (AWS + App Store + SEO + Final Documentation)	30
13	12	Documentation + Final Report + Diagrams	30
14	13	Final Revisions, Buffer, Submission	-

6. 2 Gantt Chart

Activity	Weeks																	
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Prototyping Data Setup (Kaggle, Meteostat, OpenFlights)									Midterm Exam Week									Exam Week
Baseline Prototype Model (XGBoost classifier & regressor)																		
Prototype Frontend (Basic Web UI for predictions)																		
Advanced Model – Phase 1 (LSTM/Transformer & Graph POC)																		
Advanced Model – Phase 1 (LSTM/Transformer & Graph POC)																		
Contingency / Buffer (EDA, fixes, reviews)																		

Figure 6.1: SEMESTER 7 (Fall 2025)

Activity	Weeks																	
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Advanced Model – Phase 2 (Hybrid integration & tuning)									Midterm Exam Week									Exam Week
Backend API Development (FastAPI, Redis caching, PostgreSQL)																		
Full Web Frontend (Next.js + Tailwind + Explainable AI UI)																		
iOS App Development (SwiftUI + notifications, saved flights)																		
Hosting & Deployment (AWS + App Store + SEO + Final Documentation)																		
Documentation + Final Report + Diagrams																		

Figure 6.2: SEMESTER 2 (Spring 2026)